Ollama

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> **Demo Environment**

Development board: Jetson Orin series mainboard

Ollama is an open source tool that aims to simplify the deployment and running of large language models, so that users can use high quality language models in a local environment.

1. Large language models (LLM)

Large language models (Large Language Models, referred to as LLM) are a class of advanced text generation systems based on artificial intelligence technology. Their main characteristic is that they can learn and understand human language from large-scale training data, and generate natural and fluent text.

2. Offline models and 4GB compatibility

Before downloading a model, choose the version according to the memory of the development board. The table below summarises the models listed in the offline model tutorials of this chapter, and adds the Qwen3-VL used by the visual understanding cases of chapter 11. The compatibility status applies to **the Jetson Orin Nano 4GB and the configuration of this course**; it is not a general conclusion for all the quantisations, context lengths and ways of running that model.

The table below marks "Supported" or "Not supported" according to the compatibility conclusion of this course; even a supported model still needs memory to be reserved for the system and the other running components. The download size of a model is not the same as the memory it needs at runtime; inference also needs the context cache, the vision components and the system memory.

Model family and tutorial	Model version	Main use	Orin Nano 4GB
Llama 3.2	`llama3.2:1b`	Text conversation	Supported
Llama 3.2	`llama3.2:3b`	Text conversation	Supported
Qwen3	`qwen3:0.6b`	Text conversation	Supported
Qwen3	`qwen3:1.7b`	Text conversation	Supported
Qwen3	`qwen3:4b`	Text conversation	Supported

Model family and tutorial	Model version	Main use	Orin Nano 4GB
Qwen3	`qwen3:8b`	Text conversation	**Not supported**
Phi-4-mini	`phi4-mini:3.8b`	Text conversation, reasoning	**Not supported**
DeepSeek-R1	`deepseek-r1:1.5b`	Text conversation, reasoning	Supported
DeepSeek-R1	`deepseek-r1:7b`	Text conversation, reasoning	**Not supported**
DeepSeek-R1	`deepseek-r1:8b`	Text conversation, reasoning	**Not supported**
Qwen2.5-VL	`qwen2.5vl:3b`	Image understanding, text conversation	**Not supported**
Qwen2.5-VL	`qwen2.5vl:7b`	Image understanding, text conversation	**Not supported**
Qwen3-VL	`qwen3-vl:2b`	Image understanding, text conversation	**Not supported (multimodal cases of this course)**
Gemma 3	`gemma3:1b`	Text conversation	Supported
Gemma 3	`gemma3:4b`	Image understanding, text conversation	**Not supported**
LLaVA	`llava:7b`	Image understanding, text conversation	**Not supported**
MiniCPM-V	`minicpm-v:8b`	Image understanding, text conversation	**Not supported**
LocateAnything	LocateAnything-3B Q4_K_M + BF16 mmproj	Image object grounding	**Not supported**

> **Limitation of the course:** the range of the small text models in the table above does not mean that the complete multimodal applications can run. The six cases of chapter 11 "Multimodal large model application (online/offline large model + text/voice interaction)" do not support the 4GB configuration, so use a Jetson Orin Nano 8GB. The fact that a model itself can run does not mean that it supports the image input or the tool calling that the cases require.

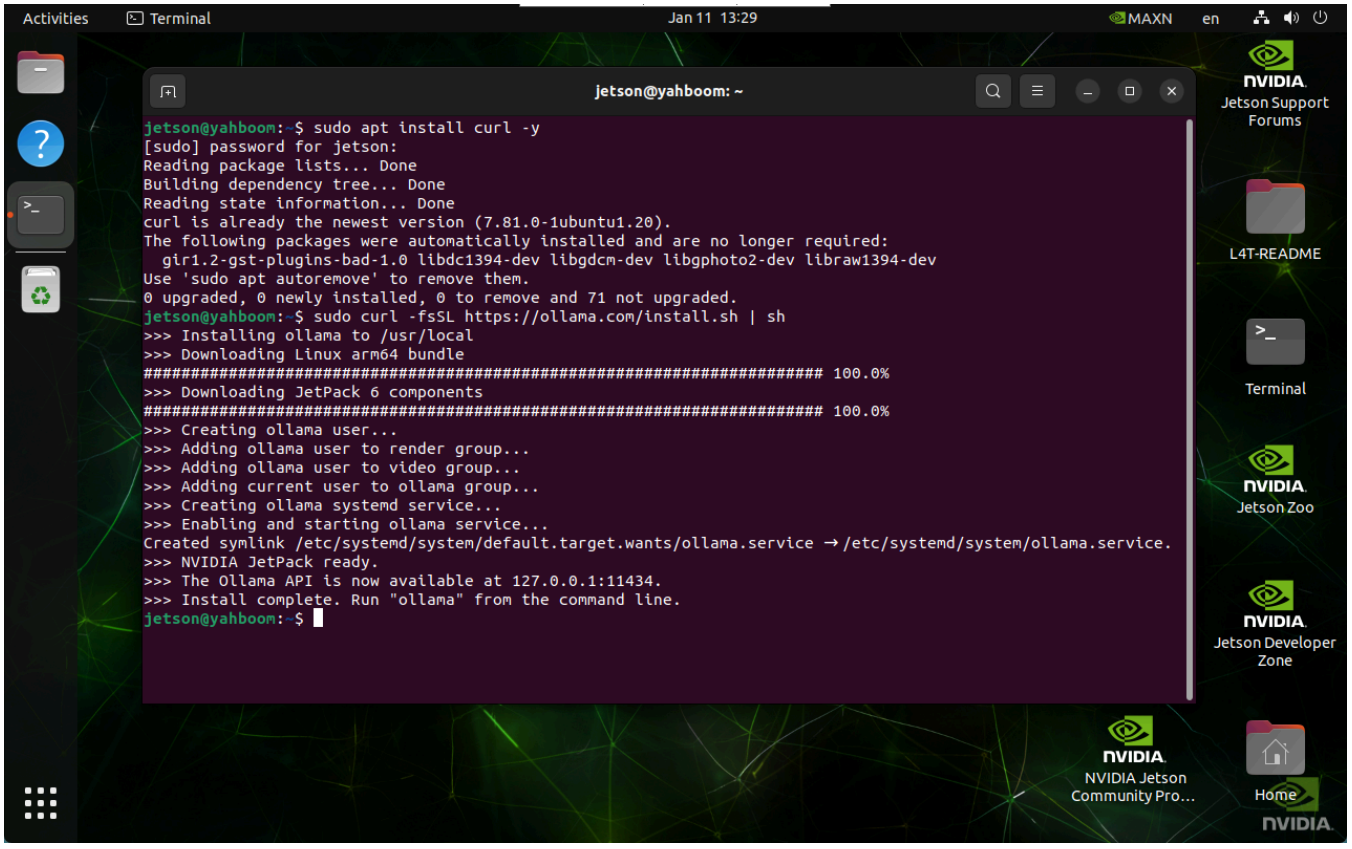
3. Installing Ollama

The tutorial demonstrates installing Ollama with a script on the Jetson Orin series mainboard. It is already installed on the factory system, so you can skip this section.

Note: a failure to pull with ollama is normal; you can try several times, or pull it through a proxy. You have to look up the proxy method yourself online

Script installation

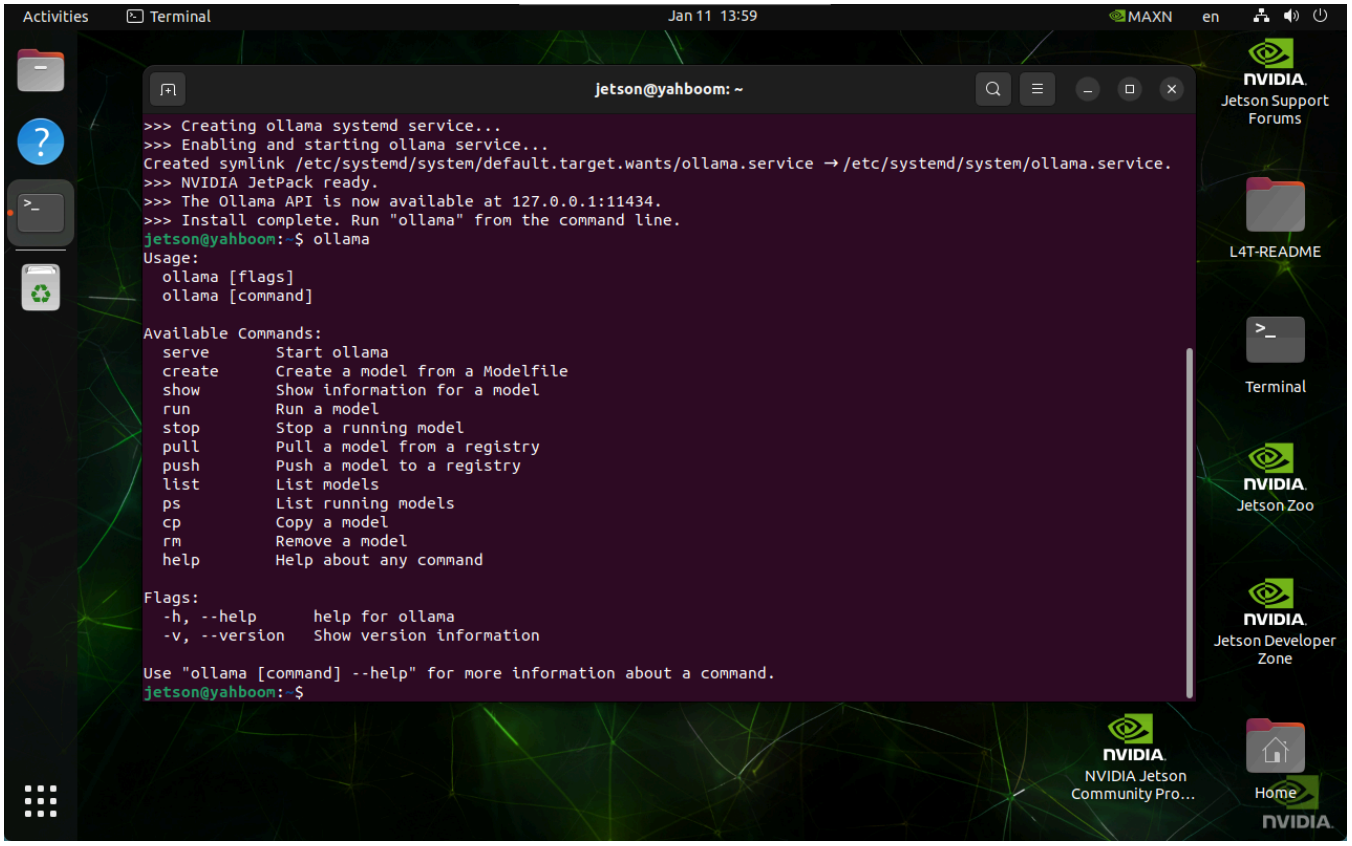
```
sudo apt install curl -y
sudo curl -fsSL https://ollama.com/install.sh | sh
```



The whole installation takes a fairly long time, so be patient!

4. Using Ollama

Entering ollama in the terminal shows the following help:



Command	Description
ollama serve	Start ollama
ollama create	Create a model from a model file
ollama show	Show the information of a model
ollama run	Run a model
ollama pull	Pull a model from the registry
ollama push	Push a model to the registry
ollama list	List the models
ollama ps	List the running models
ollama cp	Copy a model
ollama rm	Remove a model
ollama help	Get the help information of any command

5. Uninstalling Ollama

- Remove the service

```
sudo systemctl stop ollama
sudo systemctl disable ollama
sudo rm /etc/systemd/system/ollama.service
```

- Remove the files

```
sudo rm $(which ollama)
```

- Remove the models, the service user and the group

```
sudo rm -r /usr/share/ollama
sudo userdel ollama
sudo groupdel ollama
```

References

> **Ollama**

Official website: https://ollama.com/

GitHub: https://github.com/ollama/ollama